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United States Patent	12401245
Kind Code	B2
Date of Patent	August 26, 2025
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Circuit board for stator windings having integral securing structures

Abstract

A stator of an electric motor includes a stator core having a plurality of teeth. A printed circuit board (PCB) is fixed with respect to the stator and includes a plurality of soldering structures defined within a surface of the printed circuit board. A winding set is disposed on the teeth. The winding set includes at least one wire that is wound around the teeth to at least partially define stator poles. Each wire of the winding set includes opposing ends that are attached to the printed circuit board at a dedicated soldering structure of the plurality of soldering structures.

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Appl. No.:	17/789547
Filed (or PCT Filed):	December 10, 2020
PCT No.:	PCT/IB2020/061783
PCT Pub. No.:	WO2021/140381
PCT Pub. Date:	July 15, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230032339 A1	Feb. 02, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: H02K3/12 (20060101); H02K1/16 (20060101); H05K1/18 (20060101)

U.S. Cl.:

CPC H02K3/12 (20130101); H02K1/16 (20130101); H05K1/181 (20130101);
H05K2201/1009 (20130101)

Field of Classification Search

CPC: H02K (3/12); H02K (1/16); H02K (2203/09); H02K (2203/12); H02K (11/33); H05K
(1/181); H05K (2201/1009); H05K (2201/09163); H05K (1/117); H05K (2201/10287);
H05K (3/3426); Y02P (70/50)

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention generally relates to electric motors, and more specifically, securing structures for maintaining a position of a winding for a stator of an electric motor.

BACKGROUND OF THE INVENTION

(2) Conventional electric motors typically include a stator having a winding that extends around teeth of a stator core. These wires are connected to a controller via a printed circuit board. The entire assembly is then overmolded or encased within a plastic material to form the structure of the stator for the electric motor. These conventional electric motors contain an intermediate conductive element between magnet wire and the printed circuit board controller. These intermediate conductive elements, commonly referred to as terminals, require specific components, such as the terminal itself, and manufacturing processes to electrically connect the terminal to the magnet wire. Added manufacturing processes are also needed to electrically connect the terminal to the printed circuit board.

SUMMARY OF THE INVENTION

(3) According to one aspect of the present invention, a stator of an electric motor includes a stator core having a plurality of teeth. A printed circuit board (PCB) is fixed with respect to the stator and includes a plurality of soldering structures defined within a surface of the printed circuit board. A winding set is disposed on the teeth. The winding set includes at least one wire that is wound around the teeth to at least partially define stator poles. Each wire of the winding set includes opposing ends that are attached to the printed circuit board at a dedicated soldering structure of the plurality of soldering structures.

(4) According to another aspect of the present invention, a stator of an electric motor includes a stator core having a plurality of teeth. A printed circuit board (PCB) is attached to the stator core and having a plurality of soldering structures defined therein. A winding set is made up of at least one wire. Each wire of the at least one wire extends from a first end that is attached to a first soldering structure of the plurality of soldering structures, around at least a portion of the plurality of teeth and to a second end that is attached to a second soldering structure of the plurality of soldering structures. Each wire is a continuous section of wire that is free of intermediate terminals.

(5) According to another aspect of the present invention, an electric motor includes a printed circuit board. A wire-securing structure is positioned proximate the printed circuit board. A stator includes a plurality of poles. A wire extends from a first end attached to the printed circuit board at a first soldering pad, through a first portion of the wire-securing structure, around at least a portion of the poles of the plurality of poles, through a second portion of the wire-securing structure and to a second end attached to the printed circuit board at a second soldering pad. The wire includes a continuous and uninterrupted conductor.

(6) These and other aspects, objects, and features of the present invention will be understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a perspective view of a stator for an electric motor that incorporates an aspect of the soldering structures for attaching the windings to a printed circuit board;

(3) FIG. 2 is an enlarged perspective view of the soldering structures of FIG. 1;

(4) FIG. 3 is a perspective view of an aspect of the stator and printed circuit board and showing a configuration of the soldering structures;

(5) FIG. 4 is an exploded perspective view of the stator of FIG. 1;

(6) FIG. 5 is a side elevational view of the stator of FIG. 1 and showing the wires of the wire set placed in a soldering position; and

(7) FIG. 6 is a top plan view of the stator of FIG. 1 and showing engagement of the wires of the wire set with the dedicated soldering pads.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(8) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the invention as oriented in FIG. 1.

However, it is to be understood that the invention may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(9) As exemplified in FIGS. 1-6, reference numeral **10** generally refers to a printed circuit board (PCB) that is incorporated within an electric motor **12**. The PCB **10** is placed in a fixed position with respect to a stator **14**, such that the various controls **16** for operating components of the stator **14** can be secured thereto during operation of the electric motor **12**. According to various aspects of the device, the stator **14** for an electric motor **12** includes a stator core **18** having a plurality of teeth **20**. The PCB **10** is fixed with respect to the stator core **18** and includes a plurality of soldering structures **22** defined within the surface **24** of the PCB **10**. A winding set **26** is disposed on the teeth **20** of the stator core **18**. The winding set **26** includes at least one wire **28** that is wound around the teeth **20** of the stator core **18** to at least partially define a plurality of stator poles **30** of the stator **14**. Typically, the winding set **26** includes a plurality of wires **28** that are each wound in a predetermined configuration around the various teeth **20** of the stator core **18**. Each wire **28** of the winding set **26** includes opposing ends **32** that are attached to the PCB **10** at a dedicated soldering structure **34** of the plurality of soldering structures **22**. Using the soldering structures **34**, each of the various wires **28** can be installed on the stator **14** and the PCB **10** as a single and continuous section of wire **28**. By using a single and continuous wire **28**, intermediate terminals and other electrical junctions are not included within the wires **28** of the winding set **26**.

(10) As exemplified in FIGS. 1-6, the plurality of soldering structures **22** are defined proximate an edge **50** of the PCB **10**. Each soldering structure **22** can include a securing notch **52** that is defined within the edge surface **54** of the PCB **10**. Each securing notch **52** is typically aligned with a corresponding tinned soldering pad **56** that receives a respective end **32** of each wire **28** of the winding set **26**. The securing notch **52** and the corresponding soldering pad **56** cooperate to make up at least a portion of each respective soldering structure **22** of the PCB **10**. The number of soldering structures **22**, in certain aspects, can correspond to the number of wires **28** in the winding set **26**. It is also contemplated that the PCB **10** can include a set number of soldering structures **22** to account for a wide range of wires **28** and a variety of winding sets **26**.

(11) Referring again to FIGS. 1-6, the corresponding tinned soldering pads **56** are pre-tinned during manufacture of a PCB **10**. As the various circuitry is printed on the base material **70** of the PCB **10**, the various soldering pads **56** of the plurality of soldering structures **22** are also positioned within specific locations on the PCB **10**. Typically, the pre-tinned soldering pads **56** are pre-tinned during the surface mount technology (SMT) process of manufacturing the PCB **10**. Through this configuration, each securing notch **52** is aligned with a pre-tinned soldering pad **56** such that ends **32** of each wire **28** of the winding set **26** can be placed within a securing notch **52** and then prepared for soldering while each wire **28** is temporarily secured to the PCB **10**.

(12) It is contemplated that the use of the plurality of soldering structures **22** can assist in accurate placement of the opposing ends **32** of each of the wires **28** of the winding set **26**. This can help to prevent crossed wires **28** and other inaccuracies that can lead to short circuits and other structural and electrical defects within a stator core **18**. This configuration also provides for a feedback

mechanism that informs a manufacturer that the PCB **10** is accurately located on the stator core **18** with respect to the various wires **28** of the winding set **26**. In this manner, the securing notches **52** assist in at least temporarily securing the opposing ends **32** of each wire **28** for the winding set **26**. The various securing notches **52** are configured to extend more than 180 degrees around an insulation layer **78** of the at least one wire **28** of the winding set **26**. By extending more than 180 degrees around the insulation layer **78** of each wire **28**, each securing notch **52** provides a clamping or other similar securing engagement to secure the at least one wire **28** in a soldering position **100**. This soldering position **100** places the wires **28** of the winding set **26** in a convenient location to be stripped to expose the interior conductor **80**. Once stripped, the interior conductor **80** can be conveniently folded at the securing notch **52** to engage the pre-tinned soldering pad **56**. Once in this location, a soldering operation can occur to attach the conductor **80** of the wire **28** to the respective soldering pad **56**. These operations are assisted through the retention of the ends **32** of each wire **28** within the securing notches **52**. This retaining function allows for a hands-free system for stripping and soldering the wire **28**.

(13) These operations of placing the wires **28** within the securing notch **52**, stripping the wires **28**, folding the wires **28** and soldering the wires **28** to the soldering pad **56** can be accomplished by hand using manually operated tools, can be operated through automatic or robotic mechanisms or a combination of manual and automated processes. The soldering structures **22** provide an integral guide that positions the wires **28** during assembly and manufacture and also helps to protect the wires **28** during use.

(14) Typically, the pre-tinned material of the soldering pads **56** is a soldering paste **90** that is applied during the SMT process. These soldering pads **56** are located in alignment with the various securing notches **52** for retaining, at least temporarily, the position of the wires **28** of the winding set **26** in the soldering position **100**. Other soldering pads **56** can be located at positions around the PCB **10** for receiving other wires **28** of the winding set **26**, as well as other data and electrical connections extending from the PCB **10** to other locations of the motor **12**, controller or other similar mechanism operated in connection with the motor **12**. When the various wires **28** of the winding set **26** are soldered to the pre-tinned soldering pads **56**, various selective soldering processes can be utilized. These selective soldering processes can be implemented for providing accurate solders that minimize the use of soldering material as well as other electrical and material resources. This process also minimizes the crossing of wires **28** such that each wire **28** of the winding set **26** is accurately positioned within the PCB **10** and also soldered to the appropriate pre-tinned soldering pad **56**.

(15) Referring again to FIGS. **1-6**, the stator **14** for the electric motor **12** includes the stator core **18** having the plurality of teeth **20**. The PCB **10** is attached to the stator core **18** and includes a plurality of soldering structures **22** defined therein. The winding set **26** is made up of at least one wire **28**, where each wire **28** of the at least one wire **28** extends from a first end **130** that is attached to a first dedicated soldering structure **118** of the plurality of soldering structures **22**. Each wire **28** extends around at least a portion of the plurality of teeth **20** for the stator core **18**. A second end **132** of each wire **28** is attached to a second dedicated soldering structure **120** of the plurality of soldering structures **22**. As discussed previously, each wire **28** of the winding set **26** is a continuous section of wire **28** that is free of intermediate terminals. Accordingly, the process of installing the windings onto the stator core **18** and then attaching those windings to the PCB **10** can be performed in a less complicated procedure, when compared to those procedures that include intermediate terminals.

(16) Referring again to FIGS. **1-6**, the stator core **18** can include various caps **110** or other similar structures that may include aligning towers **112**. These aligning towers **112** are typically aligned with respective securing notches **52** of the plurality of soldering structures **22**. These aligning towers **112** can include a wire-receiving interior channel **114** that receives a wire **28** of the winding set **26**. Using these aligning towers **112**, the various wires **28** of the winding set **26** can be directed

from the stator teeth **20** of the stator core **18** and can be specifically directed to engage a dedicated securing notch **52** of a dedicated soldering structure **34**. Again, using the aligning towers **112** in combination with the plurality of soldering structures **22**, each wire **28** of the winding set **26** can be specifically directed and accurately positioned within a particular location of the PCB **10** for accomplishing a precise selective solder process. Also, the aligning towers **112**, in combination with the securing notches **52**, can operate to protect the wires **28** during use of the stator core **18**. (17) Through the use of the plurality of soldering structures **22** of the PCB **10**, which include the securing notches **52** and the pre-tinned soldering pads **56**, the at least one wire **28** of the winding set **26** can be in the form of a continuous wire **28** that is free of intermediate terminals.

Accordingly, a single continuous wire **28** can be incorporated within the stator **14** for use as a wire **28** of the winding set **26** for the stator **14**. Using this configuration, each wire **28** of the winding set **26** can be a single and continuous wire **28** that extends from a first dedicated soldering structure **118** of the PCB **10**, around various teeth **20** of the stator core **18** and then to a second dedicated soldering structure **120** of the PCB **10**. By using segments of continuous wire **28** for the winding set **26**, minimal resources can be utilized for accomplishing the various winding processes of the winding set **26**. Additionally, because no intermediary terminals are used within the wires **28** of the winding set **26**, there is less of a chance of manufacturing defects or tolerances that might affect the performance of the various components of the electric motor **12**.

(18) Within conventional electric motors, the intermediary terminal located between magnet wire and the printed circuit board controller require additional components, at least in the form of the intermediary terminal itself and other connective components. Additional manufacturing processes are also needed within these conventional motors to electrically connect the intermediary terminal with each of the magnet wire and the printed circuit board controller. This is true for every coil wound in the conventional stator. These added components and processes can be a source of strain on the magnet wires and other components of the conventional motors that can result in material fatigue or failure in the magnet wires, the solder connections and other locations.

(19) Referring again to FIGS. **1-6**, an electric motor **12** includes a PCB **10** and a wire-securing structure that is positioned proximate to the PCB **10** in the form of the soldering structures **22**. A stator **14** for the electric motor **12** includes a plurality of poles. These poles each include various wires **28** that make up a winding set **26** for the stator **14** of the electric motor **12**. A wire **28** extends from a first end **130**, which is attached to the PCB **10** at a soldering pad **56** of the first dedicated soldering structure **118**. The wire **28** then extends through a first portion of a wire-securing structure, typically in the form of a securing notch **52**. The wire **28** extends around at least a portion of the poles of the plurality of poles to form the various windings. The wire **28** extends to a second portion of the wire-securing structure that includes a second securing notch **52**. As discussed herein, each of the securing notches **52** are defined within an edge **50** of the body of the PCB **10**. The wire **28** includes a second end **132** that is attached to the PCB **10** at a soldering pad **56** of the second dedicated soldering structure **120**. The wire **28** includes a continuous and uninterrupted conductor **80** that is free of intermediate terminals.

(20) Referring again to FIGS. **1-6**, first and second soldering pads **56** are positioned adjacent to respective securing notches **52** of the wire-securing structure. These securing notches **52** are defined within an edge **50** of the PCB **10**. In operation, the respective securing notches **52** are configured to selectively fix the first and second ends **130**, **132** of the wire **28** in a soldering position **100**, in relation to the PCB **10**. This soldering position **100** (shown in FIG. **5**) is characterized by the first and second ends **130**, **132** of the wire **28** being oriented to allow for stripping and soldering of the wire **28** while the first and second ends **130**, **132** of the wire **28** are in the soldering position **100**. Stated another way, the soldering position **100** places each of the first and second ends **130**, **132** of the wire **28** in a generally perpendicular configuration with respect to the PCB **10**. From this position, the first and second ends **130**, **132** of the wire **28** are easily accessible to strip away the insulation layer **78** for the wire **28** and expose the interior conductor **80**.

This conductor **80** can become easily bent and attached to the respective soldering pad **56** to complete the installation of the wire **28** for the winding set **26**.

(21) Referring again to FIGS. **1-6**, the use of the continuous wires **28** of the winding set **26**, as well as the securing notches **52** and pre-tinned soldering pads **56**, creates strain relief features. These strain relief features are used to limit the material stresses experienced within the wires **28**, the PCB **10** and other components of the motor **12** during manufacture and use. The improved construction of the stator **14** for the motor **12** via a novel means of connection of the wires **28** of the stator core **18** directly to the PCB **10** without an intermediate terminal creates the integrated strain relief within these components. As discussed herein, the features located within and near the edge **50** of the PCB **10**, typically in the form of the securing notches **52** and pre-tinned soldering pads **56**, create the strain relief features. These strain relief features serve to clamp on the insulation layer **78** of the wire **28**, while the mechanically stripped and exposed interior conductor **80** of the wire **28** is soldered to pre-tinned solder pads **56** on the PCB **10**. The securing notches **52** also provide a guide around which the wire **28** can be bent for soldering the conductor **80** of the wire **28** to the soldering pad **56**.

(22) Utilizing the plurality of soldering structures **22** in the PCB **10**, the securing notches **52** and pre-tinned soldering pads **56** provide an integral strain relief feature of the PCB **10** that minimizes stresses that are placed upon the various wires **28** of the winding set **26** during manufacture of the electric motor **12**. These strain relief features minimize breakage and other damage that may occur to the solder joint or fatiguing of the wires **28** of the winding set **26**, and other components of the electric motor **12**.

(23) The PCB **10** incorporating the soldering structures **22** can be utilized within a wide variety of electric motors. Such motors can include, but are not limited to, brushless DC motors, synchronous A/C motors, switched reluctance, synchronous reluctance and other similar motor applications that utilize a stator **14** and a winding set **26**.

(24) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present invention, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A stator of an electric motor, the stator comprising: a stator core having a plurality of teeth; a printed circuit board (PCB) that is fixed with respect to the stator and includes a plurality of soldering structures defined within a surface of the PCB; and a winding set that is disposed on the plurality of teeth, wherein the winding set includes at least one wire that is wound around the plurality of teeth to at least partially define stator poles, wherein each wire of the winding set includes opposing ends that are attached to the PCB at a dedicated soldering structure of the plurality of soldering structures, wherein the plurality of soldering structures are defined at an edge of the PCB and wherein each soldering structure includes a securing notch that is defined within the edge of the PCB, and wherein the stator core includes aligning towers that are aligned with a respective securing notch of the plurality of soldering structures.
2. The stator of claim 1, wherein each securing notch is aligned with a corresponding soldering pad, wherein each corresponding soldering pad is pre-tinned.
3. The stator of claim 2, wherein each securing notch of the plurality of soldering structures is configured to position the at least one wire in a soldering position so that the at least one wire can be stripped and soldered to the corresponding soldering pad, while the at least one wire is retained in a respective securing notch.
4. The stator of claim 3, wherein each securing notch is configured to extend more than 180 degrees around an insulation layer of the at least one wire to secure the at least one wire in the

soldering position.

5. The stator of claim 2, wherein the corresponding soldering pads are pre-tinned during a Surface Mount Technology (SMT) Process.

6. The stator of claim 1, wherein each aligning tower includes a wire-receiving interior channel that positions the at least one wire in alignment with the respective securing notch, and wherein the at least one wire is a continuous wire that is free of intermediary terminals.

7. The stator of claim 6, wherein the at least one wire extends from a first dedicated soldering structure of the PCB, around the plurality of teeth of the stator core and to a second dedicated soldering structure of the PCB.

8. A stator of an electric motor, the stator comprising: a stator core having a plurality of teeth; a printed circuit board (PCB) attached to the stator core and having a plurality of soldering structures defined therein; and a winding set made up of at least one wire, wherein each wire of the at least one wire extends from a first end that is attached to a first soldering structure of the plurality of soldering structures, around at least a portion of the plurality of teeth and to a second end that is attached to a second soldering structure of the plurality of soldering structures, wherein each wire is a continuous section of wire that is free of intermediate terminals.

9. The stator of claim 8, wherein the plurality of soldering structures are defined at an edge of the PCB and wherein each soldering structure includes a securing notch that is defined within the edge of the PCB.

10. The stator of claim 9, wherein each securing notch is aligned with a corresponding soldering pad.

11. The stator of claim 10, wherein the respective securing notch is configured to selectively fix a portion of the at least one wire in a soldering position in relation to the corresponding soldering pad, wherein the soldering position is characterized by the at least one wire being located to allow for stripping and soldering while the at least one wire is in the soldering position.

12. The stator of claim 10, wherein each of the corresponding soldering pads are incorporated within respective soldering structures of the plurality of soldering structures.

13. The stator of claim 10, wherein each corresponding soldering pad is pre-tinned, and wherein each corresponding soldering pad is pre-tinned during a Surface Mount Technology (SMT) Process.

14. The stator of claim 8, wherein the at least one wire extends from a first dedicated soldering structure of the PCB, around the plurality of teeth of the stator core and to a second dedicated soldering structure of the PCB.

15. An electric motor comprising: a printed circuit board (PCB); a wire-securing structure that is positioned proximate the PCB; a stator having a plurality of poles; and a wire that extends from a first end attached to the PCB at a first soldering pad, through a first portion of the wire-securing structure, around at least a portion of the poles of the plurality of poles, through a second portion of the wire-securing structure and to a second end attached to the PCB at a second soldering pad, wherein the wire includes a continuous and uninterrupted conductor.

16. The stator of claim 15, wherein the wire-securing structure is at least partially defined within an edge of the PCB.

17. The stator of claim 15, wherein each of the first and second soldering pads are positioned adjacent respective securing notches of the wire-securing structure, the respective securing notches being defined within an edge of the PCB, wherein the first and second soldering pads are pre-tinned during a Surface Mount Technology (SMT) Process.

18. The stator of claim 17, wherein the wire-securing structure further includes aligning towers that are positioned in alignment with the respective securing notches of the PCB, wherein each aligning tower includes an interior channel that positions the wire in alignment with the respective securing notches.

19. The stator of claim 17, wherein the respective securing notches are configured to selectively fix

the first and second ends of the wire in a soldering position in relation to the PCB, wherein the soldering position is characterized by the first and second ends of the wire being oriented to allow for stripping and soldering of the wire while the first and second ends of the wire are in the soldering position.

20. The stator of claim 17, wherein the respective securing notches are configured to extend more than 180 degrees around an insulation layer of the wire to secure the wire in a soldering position.
